Identifying the Effects of Forced Responses in Soil Moisture–Precipitation Coupling Mechanisms

Mengxue Zhang[1], Feini Huang[1], Andrei Gavrilov[1], Nathan Mankovich[1], Miguel-Ángel Fernández-Torres[2], Gustau Camps-Valls[1] [1]Universitat de València, Image Processing Laboratory, València. Spain [2]Department of Signal Theory and Communications, Universidad Carlos III de Madrid (UC3M), Madrid, Spain

EGU General Assembly 2026
#EGU26
Abstract EGU26-8630

1. Problem Statement

- Current attribution frameworks neglect nonlinear causal interactions, blurring the fingerprint of external forcing on couplings.
- We integrate causal discovery with statistical attribution to isolate how much of the soil moisture–precipitation feedback is forced versus internal.
- We reveal anthropogenic warming is shifting the global SM–P coupling.

3. Datasets

Variable	Source
Soil Moisture (SM)	ERA5-Land, daily, 1981-2020
Precipitation (P)	ERA5, daily, 1981-2020
Latent Heat Flux (LHF)	ERA5, daily, 1981-2020
Sensible Heat Flux (SHF)	ERA5, daily, 1981-2020
Air Temp. (T2M)	ERA5, daily, 1981-2020
Vertical Velocity (ω500)	ERA5, daily, 1981-2020
Surface Pressure (SP)	ERA5, daily, 1981-2020
Sea Surface Temperature (SST)	ERA5, daily, 1981-2020
Surface Solar Radiation Downwards (SSRD)	ERA5, daily, 1981-2020

Table 1. All datasets are rescaled to a 2.5° global grid. The data processing includes deseason, warm-season slice, and detrend.

4.1 Spatio-temporal Restructuring of LHF -> P

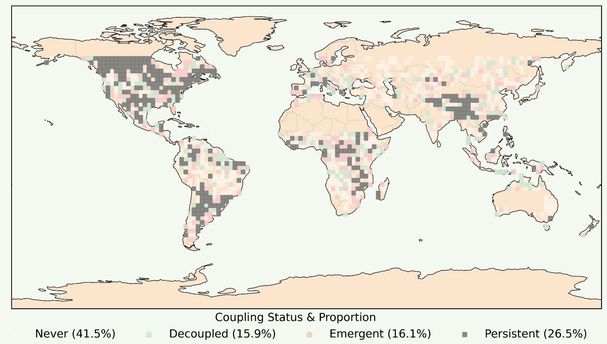

Fig. 2 illustrates the global spatial migration of "causal salient regions" of LHF -> P (Never, Persistent, Emergent, Decoupled) over 1981–2020.

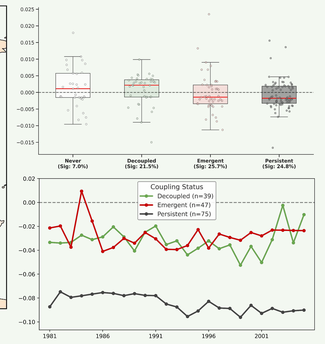

Fig. 3 Temporal evolution of causal salient regions, the box plot reveals the spread of trends.

2. Methodology

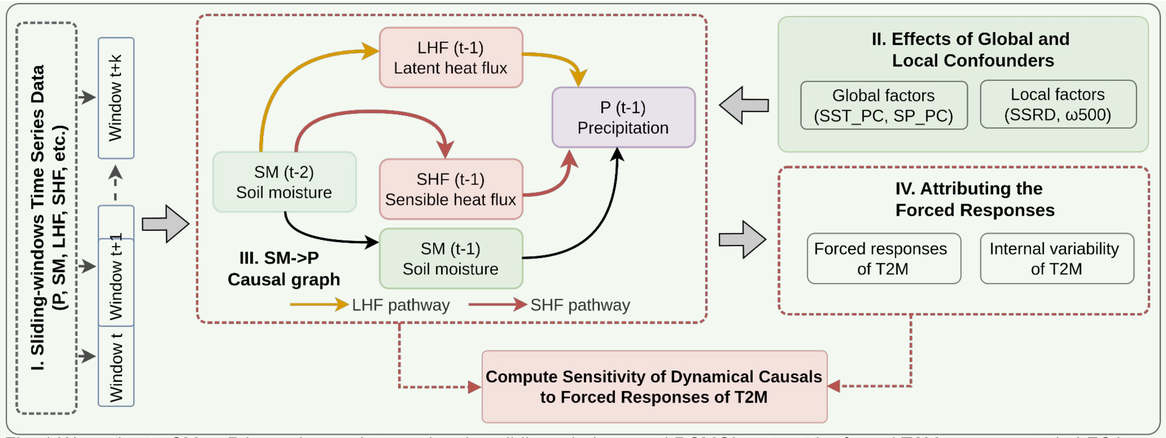

Fig. 1 We estimate SM -> P lagged causal strength using sliding-windows and PCMCI, extract the forced T2M component via LFCA, then regress causal links onto the forced signal to quantify forced causal sensitivity.

Key Takeaway

LHF -> P causal salient regions shift over ~32% of global land.

Forced temperature rather than climate variability explains this restructuring.

Future: epistemic uncertainty and Earth System Model evaluations.

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4.2 T2M Forcing Responses and the Restructuring Effects

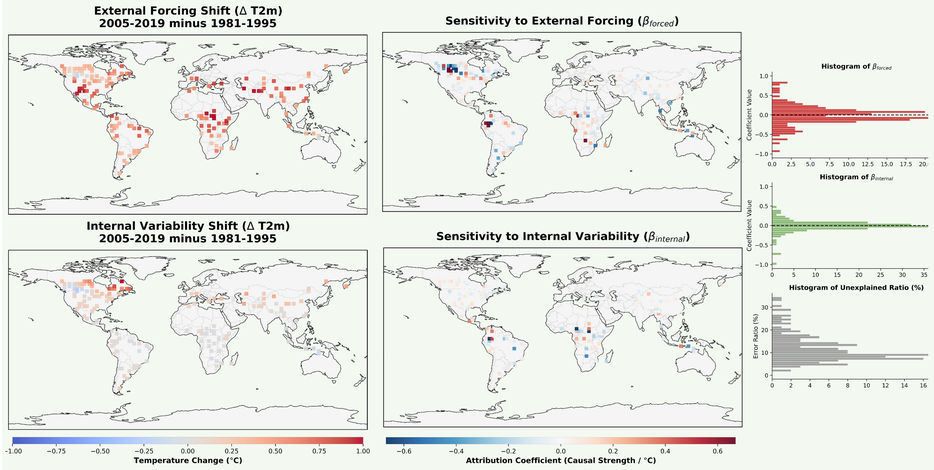

Fig. 4. Contrasting forced vs. internal T2M changes of 1981–1995 and 2005–2019.

Fig. 5 shows spatial fingerprints of forced causal sensitivity, peeking over central North America, northern South America, and central South Africa.

References

[Seneviratne, Sonia I., et al, 2010] Investigating soil moisture–climate interactions in a changing climate: A review. Earth-Sci. Rev.
[Runge, Jakob, et al, 2019] Detecting and quantifying causal associations in large nonlinear time series datasets. Sci. Adv.
[Sun, Jing, et al, 2025] Causal pathways underlying global soil moisture–precipitation coupling. Nat. Commun.
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[Qing, Yamin, et al, 2023] Soil moisture−atmosphere feedbacks have triggered the shifts from drought to pluvial conditions since 1980. Commun. Earth Environ.
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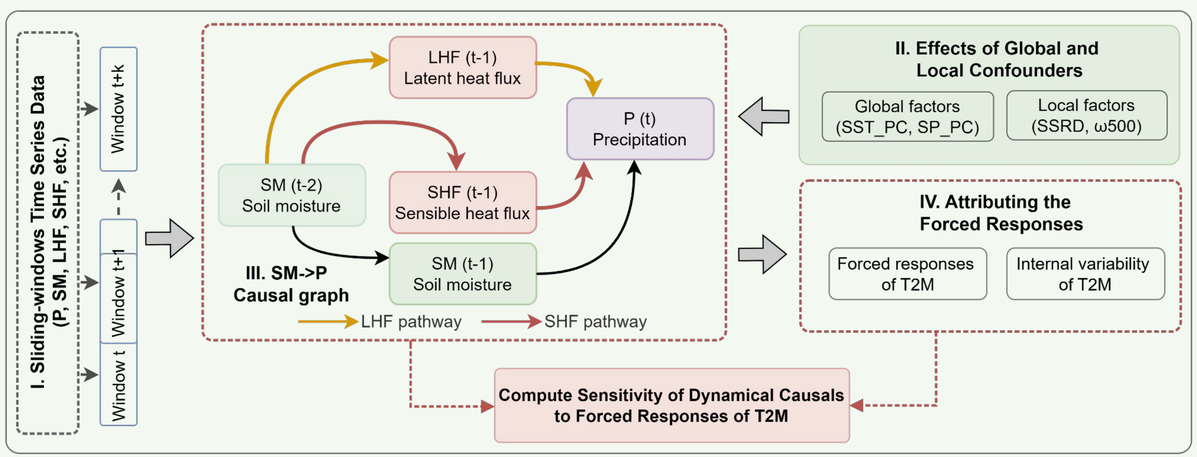

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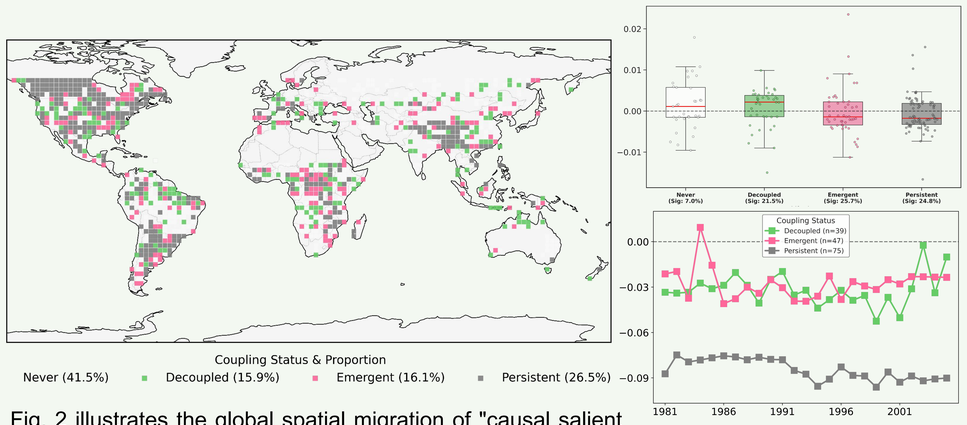

Fig. 2 illustrates the global spatial migration of "causal salient regions" of LHF -> P (Never, Persistent, Emergent, Decoupled) over 1981–2020.

Fig. 3. Temporal trends and evolutions of regions (Sig.)

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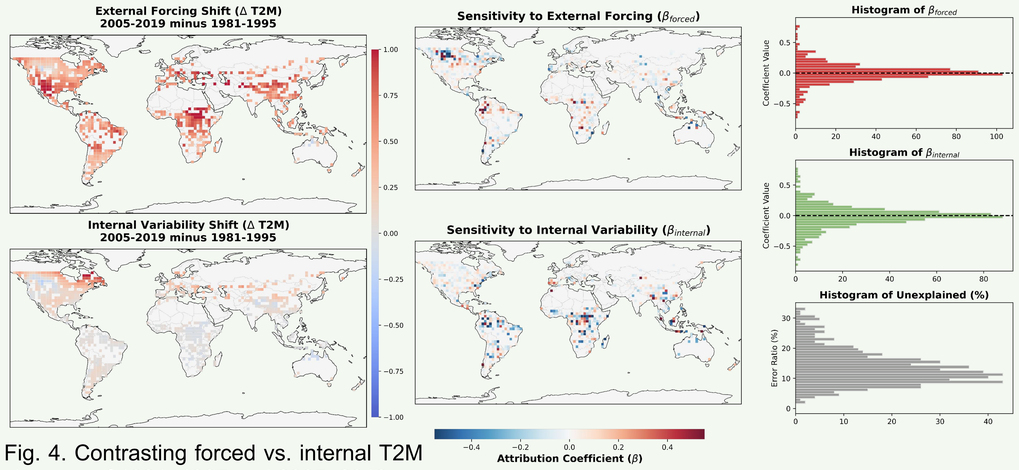

Fig. 4. Contrasting forced vs. internal T2M changes of 1981–1995 and 2005–2019.

Fig. 5 shows spatial fingerprints of forced causal sensitivity.

Key Takeaway

LHF -> P causal salient regions shift ~32% of global land.

Forced temp instead of climate variability explains North America shifts.

Does causal prefiltering bias fingerprints versus all pixels? How balance the reliability and the number?

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References

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